

Python's difflib SequenceMatcher speed up

Asked 7 years, 8 months ago Modified 1 year, 1 month ago Viewed 5k times


4


I'm using difflib SequenceMatcher (ratio() method) to define similarity between text files. While difflib is relatively fast to compare a small set of text files e.g. 10 files of 70 kb on average comparing to each other (46 comparisons) takes about 80 seconds.

The issue here is that i have a collection of 3000 txt files (75 kb on average), a raw estimation on how much time SequenceMatcher needs to complete the comparison job is 80 days!

I tried "real_quick_ratio()" and "quick_ratio()" methods, but they don't fit to our needs.

Is there any way to speed up the comparison process? If not, is there any other faster method to do such a task? Even if it is not in Python.

[python](#) [performance](#) [python-2.7](#) [difflib](#)

Share Follow

edited May 14, 2016 at 13:06



pppery
3,551 19 28 41

asked Sep 5, 2014 at 7:47

user734094

3 Answers

Sorted by:

Highest score (default) 

The issue you're finding is very common, since `difflib` is not optimized. Here are some tricks I've found over the years while developing a tool that compares HTML documents.



Files fit in memory

Create two lists, containing the lines from each file. Then call `difflib.SequenceMatcher` with the lists as parameters. The `SequenceMatcher` knows how to handle lists, and the process will be much faster since it is done on a line by line basis instead of char by char. This might reduce the precision.

Take a look at [fuzzy_string_cmp.py](#) and [diff.py](#) to see how I'm doing exactly this.

Alternative

There is a great library called [diff_match_patch](#) which is available in pypi. The library will perform **fast** diffs between two strings and return the changes (line added, line equal, line removed).

By leveraging [diff_match_patch](#) you should be able to create your own `dmp_quick_ratio` function.

In [diff.py](#) you can see how I'm using the library to get inspiration for creating `dmp_quick_ratio`.

My tests showed that using [diff_match_patch](#) was 20 times faster than Python's `difflib`.

Share Follow

edited Jun 13, 2018 at 21:21

answered May 9, 2018 at 12:01



[andres.riancho](#)

1,126 13 18

There is a C implementation of `difflib.SequenceMatcher` , [cdifflib](#).

1 Replace the `SequenceMatcher` and all `difflib` operations will be about 4x faster

```
from cdifflib import CSequenceMatcher
import difflib
difflib.SequenceMatcher = CSequenceMatcher
```

Share Follow

answered Mar 18, 2021 at 15:30



[xsdf](#)

385 2 14

You can get a small speedup using pypy

-7 <http://pypy.org/>

Share Follow

answered Jun 30, 2015 at 7:32



[ark](#)

648 2 7 27

2 This recommendation is too generic. – [andres.riancho](#) May 7, 2018 at 19:22